CHEM 427/527 · SESSION 8 · TUESDAY, OCTOBER 27, 2026

Noise in chemical measurements

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Drafted with assistance by Claude Code. Reviewed and revised by Matt.

LEARNING OBJECTIVES

By the end of today you can

1. Show from a real ICP-MS count trace that ion counts scatter with a variance close to their mean. Find the window length where a count four times longer stops halving the relative noise.
2. Compute the spike and the blank method detection limit (MDL) by 40 CFR 136 Appendix B at each integration time, and report the greater.
3. Choose an integration time and a background treatment from the MDL each gives.
4. Report each home's lead as a number or as zero by the MDL. Check an assistant's MDL against Appendix B, and choose your follow-on from what the check finds.

FIRST, EVERYONE PULLS

Get session 8 with git pull

```
git pull
```

S08-noise appears, with

- **data/ag_trace_1ms.csv**: a real silver count trace, 1 ms per reading
- **data/method_study.csv**: one lab's MDL study
- **data/labs/**: 36 labs, one for each of you

Session 7 is due today

- Upload it by 1:30 pm.
- Restart and Run All before you upload.

Nothing to download today: the silver trace, the method study, and every lab are in the repository.

FROM ONE ABSORBANCE TO MANY COUNTS

What do ten readings of one solution look like?

data/ag_trace_1ms.csv: a header line, then one reading per line

```
107Ag, counts
```

then 6, 4, 3, 4, 2, 1, 4, 1, 6, 1, and on for 60 s

A reading

The silver ions counted in 1 ms: a whole number.

The trace

59,999 readings over 60.0 s, with a mean of 2.800 counts.

The question

How far does one reading stray from the mean, and what does counting longer do to that?

THE DECISION

Is a home's lead a number, or zero?

The rule

40 CFR 141.89: an MDL for lead of at most 1 µg/L (1 ppb). A result below it is reported as zero.

MDL set too low

Results the lab cannot tell from its own blanks go into a home's record as lead.

MDL set too high

Homes with lead are recorded as zero.

Real: data/ag_trace_1ms.csv

Dissolved silver at 0.1 µg/L, read in 1 ms windows for 60 s, from Abad Álvaro, Bolea Morales, and Laborda García (2026).

Simulated: the lead studies

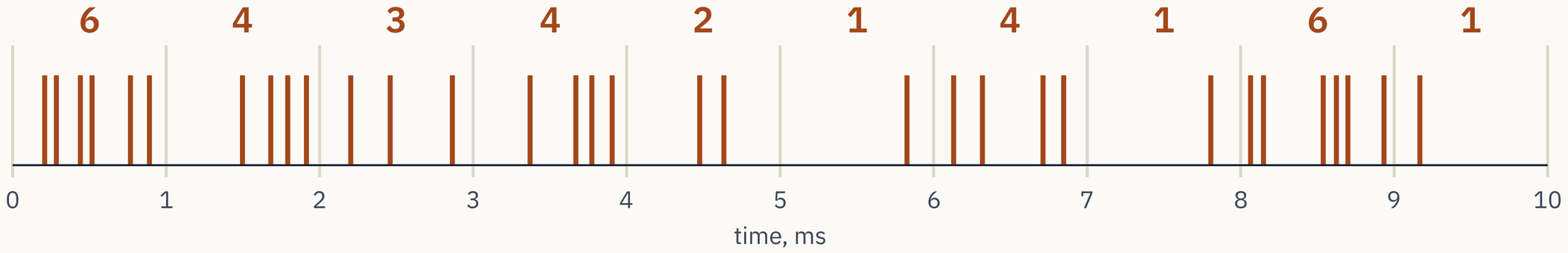
From a stated counting model, at a sensitivity well below a modern instrument's, so that counting noise shows.

The lab manager's two choices, the integration time and the background correction, set the MDL.

BEFORE THE DATA: WHERE THE SCATTER COMES FROM

Ions arrive one at a time, at random

Each mark is one ion reaching the detector; each number is the ions in that 1 ms window.



Steady on average

The solution does not change, so ions arrive at a steady average rate: 2.8 per ms here.

Random one by one

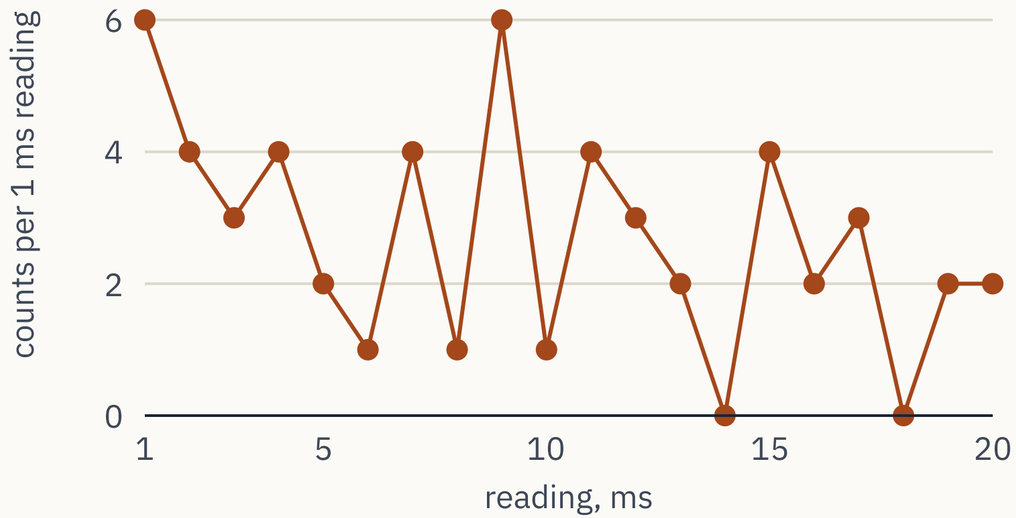
Each ion arrives independently of the others, so no window is promised exactly 2.8.

Shot noise

A window that averages N ions scatters by about √N. It comes from counting, not from a fault.

LIVE, IN A SCRATCH NOTEBOOK: THE FIRST 20 MS

One solution, twenty different counts




```
counts = np.loadtxt("data/ag_trace_1ms.csv",
skiprows=1)
ax.plot(counts[:20], "o-")
```

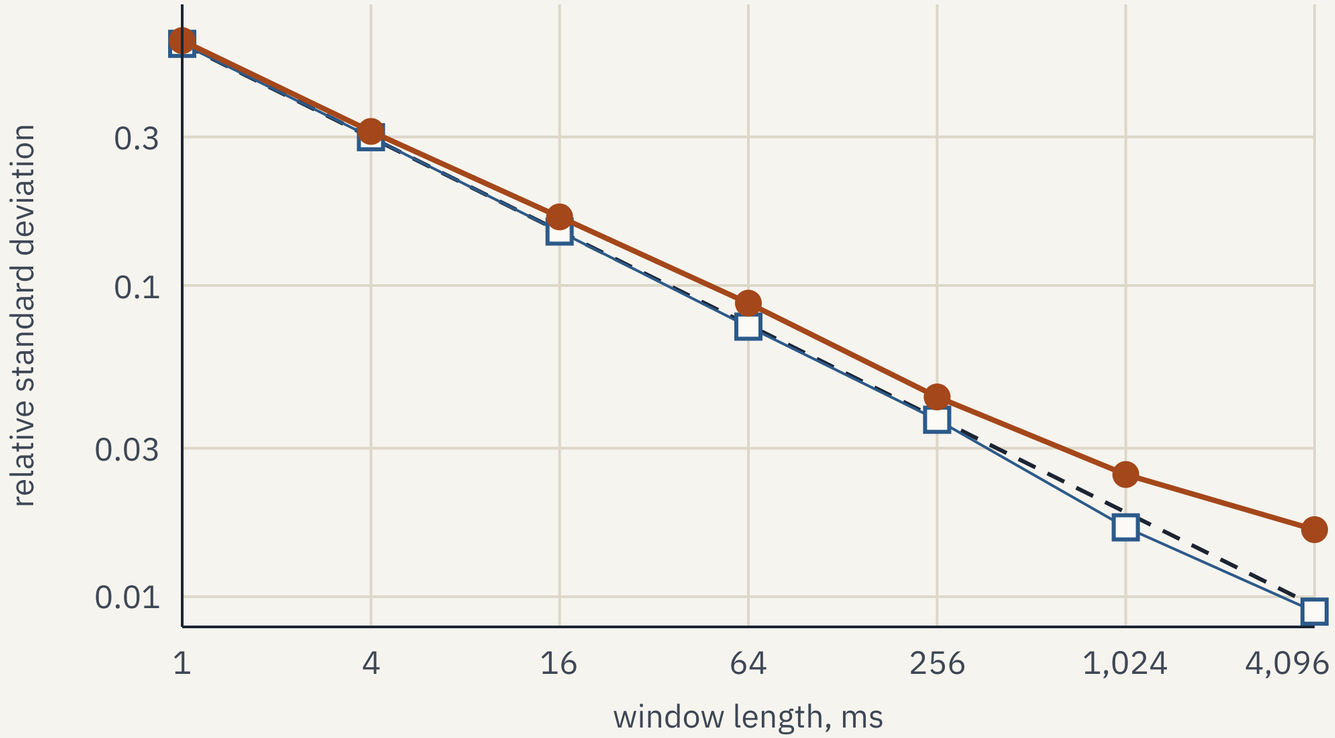
Then sum the readings into windows with reshape:

Window	Mean counts	Relative SD	1/√N
4 ms	11.2	0.312	0.299
16 ms	44.8	0.166	0.149
64 ms	179	0.0878	0.0747

Each window four times longer roughly halves the relative noise.

SECTION 1: THE TRACE SUMMED INTO LONGER WINDOWS

Where counting longer stops paying



Both axes are on log scales.

Measured, the silver trace

1/√N, counting alone

Poisson, seeded simulation

To 16 ms the trace follows 1/√N: 0.166 against 0.149. At 1.02 s it is 0.0247 against 0.0187.

The extra noise comes from the plasma and the sample introduction, which counting longer does not average away.

NOTES SECTION 3, WORKED ON THE METHOD STUDY AT 0.4 S

The MDL is the greater of two

Spike MDL

t times the standard deviation of the seven spikes.

3.143 × 0.0949 = **0.298 µg/L**

Blank MDL

The seven blanks' mean plus t times their standard deviation.

0.209 + 3.143 × 0.139
= **0.646 µg/L**

3 × SD of the blanks

The older shortcut: 3 in place of t, and no blank mean.

3 × 0.139 = **0.417 µg/L**

t = 3.143 is Student's t at 99% for six degrees of freedom, T_99 in Parameters. The spike MDL falls from 1.306 µg/L at 0.025 s to 0.298 at 0.4 s, and only to 0.281 at 1.6 s.

The MDL is 0.646 µg/L, set by the blanks. A longer count shrinks the counting noise, not the lead the blanks carry.

ASSIGNMENTS

One lab each

#	Lab
1	lab_01
2	lab_02
3	lab_03
4	lab_04
5	lab_05
6	lab_06
7	lab_07

#	Lab
8	lab_08
9	lab_09
10	lab_10
11	lab_11
12	lab_12
13	lab_13
14	lab_14

#	Lab
15	lab_15
16	lab_16
17	lab_17
18	lab_18
19	lab_19
20	lab_20

#	Lab
21	lab_21
22	lab_22
23	lab_23
24	lab_24
25	lab_25
26	lab_26

In Parameters, set **LAB** to your lab, such as LAB = "lab_01", run the Parameters cell, then the cell after TASK 1. Section 1 is the worked example; section 2 is your lab.

AI PRACTICE 5 OF 8: CORRECT EARLY, AND CLEAR CONTEXT

Correct once, or clear and restart

Live, in Claude Code on a folder with no notes to read

Seven blanks read 0.241, 0.282, 0.332, 0.270, 0.314, 0.043 and -0.020 µg/L. What is the detection limit?

A likely first answer: 3 × SD, 0.417 µg/L. Correct it once:

Use 40 CFR 136 Appendix B: the blanks' mean plus t times their SD, t = 3.143 for seven.

The second answer adds the mean, 0.209, and gives 0.646 µg/L.

Which move, from what you find

- **Right the first time.** Verify it against a value you can check.
- **One thing wrong.** Correct it in one sentence that names the fix.
- **Several wrong, or two corrections failed.** Type /, choose **Clear conversation**, and send a better first prompt.

A correction in the first exchange costs one message. A wrong answer left in the conversation pulls the next one toward it.

TASK 2, BEFORE YOUR MDL TABLE

Your own prompt, two added cells

In Claude Code, after you save the notebook

1. Write your own prompt for your lab's spike MDL, blank MDL, and the greater at each integration time. Ask for the code in the chat, its result stored as assistant_mdl.
2. Check the answer against notes section 3: t = 3.143, the blanks' mean, the greater of two.
3. Compare your prompt and answer with a neighbor's.
4. Send one follow-on, chosen from what you found.

Then add a markdown cell, filled in

```
**Prompt record**
- **My prompt:**
- **How my neighbor's differed:**
- **My follow-on:**
```

and **a code cell** with the assistant's code, its last line setting assistant_mdl.

To add a cell: hover below a cell and click **+ Markdown** or **+ Code**, or press Esc, then B.

Both cells count toward execution: a filled-in record, and code that runs.

THE JUDGMENT CELL, SECTION 4

Your lab's MDL, and one entry

(a) Your answer

- Which integration time and background treatment would you set, and what MDL would it report?
- Your spike MDL, blank MDL, and the greater; homes reported as numbers; calls the spike MDL alone would change
- What the other treatment would have cost

(b) One judgment entry

- What the assistant suggested
- What you did: changed, rejected, or verified
- Why, from a value your notebook printed

Scored 0 to 2. Your prompt record from TASK 2 is the source; your blank MDL against the assistant's is the evidence to cite.

SECTION 1 IS READ AND RUN. SECTION 2 IS YOURS.

Go, then submit

Submit

Fill in your prompt record, then Restart, Run All, save, and upload activity.ipynb by **1:30 pm Thursday, October 29**.

Every lab is in the repository, so the graders' run reads the same data yours did.

Scoring, 0 to 3

1 if it runs after a restart and holds your prompt record, **0** if not.

0 to 2 for the judgment cell.

Help

Ask Matt, Lucas, or Chris in the room.

The third hour is an office hour.